

Problem Set 1

Problem Set 1 covers Lectures 1–4: group axioms, $\mathbb{Z}_n$, D_n and S_3 , subgroups and generators, cosets and Lagrange’s theorem, homomorphisms, isomorphisms and kernels. Direct products will appear on Problem Set 2.

Due: Wednesday, 14 October 2026, at the start of the lecture.

Reading: Zee I.1–I.2; Bünemann Ch. 2; lecture notes for Weeks 1 and 2.

Justify every answer; a correct answer without reasoning earns partial credit.

Conventions

These are the conventions of the lectures and the course simulations.

- The vertices of a regular n -gon are labelled $1, 2, \dots, n$ counterclockwise.
- r is the counterclockwise rotation by $2\pi/n$, and s is the reflection in the axis through vertex 1 and the centre.
- Products are read right to left: in gh , the element h acts first. So rs means “first s , then r ”.
- A symmetry is written as a permutation of positions: $i \rightarrow j$ when the vertex at position i is carried to position j . For the triangle, $r = (123)$, $s = (23)$ and $rs = (12)$.
- Permutations compose the same way: $(12)(23)$ means first (23) , then (12) .
- D_n has the relations $r^n = s^2 = e$ and $srs = r^{-1}$, and every element has a unique normal form $r^k s^m$ with $0 \leq k < n$ and $m \in \{0, 1\}$.
- $\mathbb{Z}_n = \{0, 1, \dots, n-1\}$ under addition mod n . D_2 is the four-element group of half-turns of a box, in which every element squares to e .

The Symmetry Explorer and the Multiplication-Table Builder on the course page can check your D_4 computations in Problems 2 and 4. Use them to check your algebra, not to replace it.

Problems

1. Group or not?

- (a) Decide which of the following are groups. For each one that is not, name an axiom that fails and give an explicit counterexample. For each one that is, say whether it is Abelian.
1. $\mathbb{Z}$ with $a * b = a - b$.
 2. The odd integers under addition.
 3. The nonzero integers under multiplication.
 4. $\{1, i, -1, -i\}$ under multiplication of complex numbers.
 5. The real 2×2 matrices with determinant 1, under matrix multiplication.
 6. $\mathbb{R} \setminus \{-1\}$ with $a * b = a + b + ab$. (Hint: compute $1 + a * b$.)
- (b) Suppose every element of a group G satisfies $g^2 = e$. Show that G is Abelian.
- (c) Let G be a finite group. Show that the elements with $g \neq g^{-1}$ come in pairs, so there is an even number of them. Deduce that every group of even order contains an element of order 2, and check this in D_3 and $\mathbb{Z}_6$.

2. Working in D_4

Take the square with vertices 1, 2, 3, 4 counterclockwise. Check that, with our conventions, $r = (1234)$ and $s = (24)$.

- (a) Write all eight elements of D_4 in normal form and as permutations in cycle notation. For each reflection, describe its axis.
- (b) Show that $sr^k = r^{-k}s$ for every integer k . Use this to bring $x = sr^3sr^2sr$ to normal form, then check your answer by multiplying the permutations.
- (c) Find the order of every element of D_4 .
- (d) D_4 has exactly ten subgroups. Find them all, say which are cyclic and which are isomorphic to D_2 , and draw the subgroup lattice: each subgroup joined by a line to the subgroups directly above it that contain it. Explain why there are no others.

3. The cyclic group $\mathbb{Z}_{12}$

- (a) Find the order of every element of $\mathbb{Z}_{12}$. Guess a formula for the order of k in terms of k and 12, and check it on all twelve elements.
- (b) Which elements generate $\mathbb{Z}_{12}$?
- (c) Find all subgroups of $\mathbb{Z}_{12}$ and draw the subgroup lattice. You may use without proof that every subgroup of a cyclic group is cyclic.
- (d) For each subgroup H , list the elements k with $\langle k \rangle = H$. What do the elements that generate the same subgroup have in common?

4. Cosets and Lagrange's theorem

- (a) In D_4 let $H = \langle s \rangle = \{e, s\}$. List the left cosets gH and the right cosets Hg . Which left cosets are also right cosets?
- (b) Repeat (a) for $K = \langle r^2 \rangle$ and for $R = \langle r \rangle$. What property of r^2 explains your answer for K ?
- (c) A group G has order 20. Which orders are possible for its subgroups, and which for its elements?
- (d) A group G has order 15, and an element $g \in G$ satisfies $g^3 \neq e$ and $g^5 \neq e$. Show that G is cyclic.
- (e) H and K are subgroups of a group G with $|H| = 4$ and $|K| = 9$. Show that $H \cap K = \{e\}$.
- (f) By Lagrange's theorem the order of every element divides $|G|$. Give an example of a group G and a divisor d of $|G|$ such that G has no element of order d .

5. Homomorphisms

- (a) Each element g of D_n acts on the plane as a 2×2 matrix $M(g)$, with $M(gh) = M(g)M(h)$. With vertex 1 on the positive x -axis,

$$M(r) = \begin{pmatrix} \cos \frac{2\pi}{n} & -\sin \frac{2\pi}{n} \\ \sin \frac{2\pi}{n} & \cos \frac{2\pi}{n} \end{pmatrix}, \quad M(s) = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Define $\varphi(g) = \det M(g)$. Show that φ is a homomorphism from D_n to the group $\{1, -1\}$ under multiplication, compute $\varphi(r^k s^m)$, and find the kernel and the image of φ .

- (b) Let $U(1) = \{z \in \mathbb{C} : |z| = 1\}$ under multiplication, and let $\psi(z) = z^2$. Show that ψ is a homomorphism from $U(1)$ onto $U(1)$, find its kernel, and show that every element of $U(1)$

has exactly two preimages. More generally, show that the kernel of $z \mapsto z^n$ is isomorphic to $\mathbb{Z}_n$. (In Week 9 the map $SU(2) \rightarrow SO(3)$ will be two-to-one in the same way.)

- (c) Find all homomorphisms $\mathbb{Z}_4 \rightarrow \mathbb{Z}_6$ and all homomorphisms $\mathbb{Z}_6 \rightarrow \mathbb{Z}_4$, and give the kernel and image of each. A homomorphism from a cyclic group is fixed by where it sends a generator; check that each candidate is well defined.
- (d) For every homomorphism $\varphi: G \rightarrow G'$ you found in (a) and (c), check that $|\ker \varphi| \cdot |\operatorname{im} \varphi| = |G|$. We will see why this always holds when we meet quotient groups.

6. Isomorphic or not?

- (a) For each pair, either give an explicit isomorphism or prove that none exists. To rule one out, find a property that isomorphisms preserve, such as being Abelian or the orders of elements, that one group has and the other lacks. You may assume that each set below is a group.
1. $\mathbb{Z}_6$ and S_3 .
 2. $\mathbb{Z}_4$ and $\{1, 2, 3, 4\}$ under multiplication mod 5.
 3. $\mathbb{Z}_4$ and $\{1, 3, 5, 7\}$ under multiplication mod 8.
 4. $(\mathbb{R}, +)$ and $(\mathbb{R}_+, \cdot)$, the positive reals under multiplication.
 5. $(\mathbb{R}, +)$ and $(\mathbb{R} \setminus \{0\}, \cdot)$.
- (b) Label the elements 1, 3, 5, 7 of the group in item 3 as positions 1, 2, 3, 4. Following Cayley's theorem, write multiplication by each element as a permutation in S_4 . What do the four permutations have in common?

7. Adding velocities in special relativity

In units with $c = 1$, two collinear velocities u and v combine to

$$u \oplus v = \frac{u + v}{1 + uv}.$$

Let $G = (-1, 1)$, the possible velocities of a massive particle.

- (a) Show that $u \oplus v \in G$ whenever $u, v \in G$. (Hint: factor $1 - u \oplus v$ and $1 + u \oplus v$.)
- (b) Find the identity and the inverse of u . Is $\oplus$ commutative?
- (c) The rapidity η of a velocity u is defined by $u = \tanh \eta$. Use

$$\tanh(\alpha + \beta) = \frac{\tanh \alpha + \tanh \beta}{1 + \tanh \alpha \tanh \beta}$$

to show that $T(\eta) = \tanh \eta$ satisfies $T(\alpha + \beta) = T(\alpha) \oplus T(\beta)$. Since T is a bijection from $\mathbb{R}$ onto $(-1, 1)$, write $u = T(\alpha)$, $v = T(\beta)$, $w = T(\gamma)$ and deduce that $\oplus$ is associative without expanding $(u \oplus v) \oplus w$. Conclude that $(G, \oplus)$ is a group isomorphic to $(\mathbb{R}, +)$.

- (d) Compute $0.5 \oplus 0.5$, then check your answer by adding rapidities. (Hint: $\tanh^{-1} u = \frac{1}{2} \ln[(1+u)/(1-u)]$.)
- (e) With c restored, $u \oplus v = (u + v)/(1 + uv/c^2)$. Which group do you recover as $c \rightarrow \infty$?